

Scheduler — Black-box I/O (matches arb drawing)

One channel, DDR5. Per rank per cycle the arb emits **1 ACT + 1 CAS + 2 PRE** winners, each {vld,bg,bank}, gated by the rank-scope can. Row path -> ACT, col path -> CAS. Parametric N_BANKS(=N_BG×4), N_BG, N_RANKS.

1 INPUTS

Per-bank path candidates (PRE_BANK_ROW_PATHS / COL_PATHS ×32)

signal	width	from/to	meaning
can_act[b] / can_cas[b] / can_pre[b]	1 ea ×N_BANKS	in	per-bank legality (into row/col/pre paths)
need_act / need_cas / need_pre[b]	1 ea ×N_BANKS	in	resident packet demand
p_bit[b]	1 ×N_BANKS	in	PRE-eligible vector (need_pre & can_pre) -> pick-2-pre
is_hit[b] / age[b]	1/few ×N_BANKS	in	CAS_W row-lock / ACT_W age (weights)
open_row[b] / col_rd_wr_ptr,rob	bus ×N_BANKS	in	row (ACT path) / {col, sram ptr, rob} (CAS path) from BANK_REQ

Rank-scope gates (AND'd with the class winner)

signal	width	from/to	meaning
can_act_rank_s[r]	1 ×N_RANKS	in	rank ACT gate: tRRD_S (cross-BG) + tFAW window
can_cas_rank_s[r]	1 ×N_RANKS	in	rank CAS gate: turnaround (tWTR/tRTW) + dqFree
can_act_bgs / can_cas_bgs[g]	1 ea ×N_BG	in	same-BG spacing (tRRD_L / tCCD_L(_WR))

Maintenance flags (ME -> scheduler, ranked with demand)

signal	width	from/to	meaning
refsb_rdy / ref_force	1 ea	in	refresh owed (ldebt) / forced (&debt==7)
rfm_rdy / rfm_force	1 ea	in	rowhammer owed (RAA>=IMT) / hard (>=MMT)
zq_rdy	1	in	ZQ calibration due (tZQ)
mrr_rdy	1	in	MR-read pending (temp poll / read MR) – uses DQ (dqFree)
mrw_rdy	1	in	MR-write pending (config / freq-change)
cas_complete[b]	1 ×N_BANKS	in	bank drained + precharged (state==IDLE)
maint_cmd_type	5	in	MAINT_PRIO winner opcode (REF/RFM/ZQ/MRW/MRR)
init_done	1	in	boot done (else DFI mux on init, scheduler gated)

Control / bus state

signal	width	from/to	meaning
gear	2	in	mc:CK ratio -> bundle width (dfi_cmd × gear)
batch_dir[r]	1 ×N_RANKS	in	WLR R/W batch direction (dir bias)
op_code (cmd)	few	in	feedback_ctl -> const-router select (writeback)
last_cas_dir/rank/bg	few	in	bus state (turnaround + diff-BG packing)
dfi_alert_n / dfi_error	1 ea	in	PHY alert / error

2 OUTPUTS

Per-rank class winners (the arb result)

signal	width	from/to	meaning
act_win[r] = {vld, bg, bank}	few ×N_RANKS	out	1 ACT winner / rank (rank_act_abr & can_act_rank_s)
act_row[r]	row ×N_RANKS	out	row for the ACT winner (row_sel of the winning bank)
cas_win[r] = {vld, bg, bank}	few ×N_RANKS	out	1 CAS winner / rank (rank_cas_abr & can_cas_rank_s)
cas_col[r] = {idx, wr/rd_ptr, column}	bus ×N_RANKS	out	col + sram ptr + rob for the CAS winner (col_sel)
pre_win[r][0..1] = {vld, bg, bank}	few×2×N_RANKS	out	2 PRE winners / rank (pick-2-pre from p_bit)

Bundle to cmd_phaser

signal	width	from/to	meaning
CMD_0..3_PACKET (vld + record)	~49×4	out	{valid, cmd_type, rw, rank, bg, bank, addr, ap, bl, rob_index, sram_addr} per phase slot
two_ui / repeat	1	out	side wire -> DFI driver (1-UI vs 2-UI; @1:1 hold)

Commit / feedback

signal	width	from/to	meaning
grant_act_oh / grant_cas_oh	1 ea ×N_BANKS	out	winner one-hot -> cells; pop-on-CAS retires the packet
grant_pre_oh[0..1]	1×2 ×N_BANKS	out	the 2 PRE winners one-hot
fsm_commit (issue-edge)	pulse×class	out	sole commit: BANK_FSM state + fires timestamp writeback

signal	width	from/to	meaning
cmd_issued{r,bg,bank,cmd}	bus	out	drives next_* writeback (WE + const-router)

Maintenance handshake (to ME)

signal	width	from/to	meaning
maint_ack	1	out	the maint cmd (cmd_type) issued -> ME clears debt/RAA, advances ba

Structure (per drawing): each rank has `rank_act_abr` (-> 1 ACT winner) and `rank_cas_abr` (-> 1 CAS winner), each ANDed with its rank-scope can (`can_act_rank_s` / `can_cas_rank_s`); `PRE_PATHS` pick-2-pre off the `p_bit` vector (-> 2 PRE winners, tPPD allows 2). ACT pulls its row via `row_sel`; CAS pulls col+ptr via `col_sel`. Winners + maint (ranked by `MAINT_PRIO`) pack into `CMD_0..3` for the phaser ($1:1/1:2/1:4 = 1/2/4$ slots). The arb NEVER rejects — legality is folded into the `can_*` inputs; it only selects + weights.